

Laravel MVC + AI Booking Calendar Guide (v2)

This updated guide extends the original booking calendar design by adding availability-based date suggestions and an optional controlled social media notification bot. All features are designed with simplicity, safety, and a student-friendly budget in mind.

1. Core Booking Features (Recap)

- Document upload (PDF/DOCX)
- AI-assisted form prefill (structured JSON only)
- Facility availability checks
- Manual approval workflow

2. New Feature: Suggested Available Dates

When a requested date is unavailable, the system suggests alternative dates. This logic is deterministic and handled entirely by Laravel, not AI.

- Query future approved bookings for the same facility
- Detect gaps between bookings
- Return the next N available dates
- Display suggestions to the user for manual selection

3. Optional Feature: Controlled Social Media Event Bot

This optional feature posts automated updates to a social media page (e.g., Facebook) for visibility of ongoing and upcoming events. It must remain strictly read-only with respect to bookings.

- Posts only approved bookings
- Posts only current or future events
- No editing or deletion of posts by the bot
- Human approval required before posting (recommended)

Recommended Approach

- Use platform scheduling APIs or manual review queues
- Run posting as a queued background job
- Store post IDs for traceability

4. Safety & Compliance Notes

- Never auto-post without admin approval (strongly recommended)
- Do not scrape or automate personal accounts
- Respect platform rate limits and policies

5. Updated Data Flow Summary

Upload → Extract Text → AI JSON Mapping → Availability Check → Suggest Dates → Manual Approval → Optional Social Notification

This version keeps advanced features optional. You should fully complete the booking and availability system before enabling social media automation.